

DOUGLAS SPENCER

☎ +44 7931 920932 ✉ douglasgspencer@gmail.com  [linkedin.com/in/douglas-gerard-spencer](https://www.linkedin.com/in/douglas-gerard-spencer)

Education

Imperial College London

Starting Sep 2026

Incoming PhD Student, Statistics and Machine Learning CDT programme

London, UK

- Beginning first-year coursework and research rotations within the CDT programme.

University of Oxford

Sep 2025 – Jun 2026

Master of Science in Mathematical Sciences

Oxford, UK

- Awarded a Merit, with an overall mark of 68.
- Dissertation exploring anytime-valid prediction sets for exchangeable data sequences, supervised by Prof. Patrick Rebeschini; awarded 82.
- Modules studied: Advanced Topics in Statistical Machine Learning, Algorithmic Foundations of Learning, Bayes Methods, Introduction to Quantum Information, Random Matrix Theory, Theories of Deep Learning.

The University of Manchester

Sep 2022 – Jun 2025

Bachelor of Science in Computer Science and Mathematics

Manchester, UK

- First Class Honours, programme percentage 81%, 3rd in cohort.
- Awarded the Hilary Kahn Prize for the department's best dissertation; the work focused on conformal prediction for quantum machine learning under the supervision of Dr Michele Caprio.
- Relevant modules: Mathematical Topics in Machine Learning, Machine Learning, Probability and Statistics 2.

Cardinal Newman College

Sep 2020 – Aug 2022

A-Levels

Preston, UK

- Mathematics A*, Further Mathematics A*, Computer Science A*, Physics A*.

Research Experience & Publications

Adaptive Conformal Prediction for Quantum Machine Learning

Aug 2024 – Jun 2025

Research Project (originally undergraduate dissertation)

- Developed the dissertation into a co-authored paper, accepted to Transactions on Machine Learning Research (TMLR).
- Extended adaptive conformal inference to variational quantum circuits, using online recalibration to handle hardware-induced distribution shift.
- Developed and ran a full experimental pipeline using TorchQuantum and Qiskit on IBM quantum hardware and simulators, enabling reproduction and benchmarking of existing conformal prediction baselines.
- Applied a measure-theoretic framework to characterise the conditions under which conformal score functions are optimal under two distinct optimality criteria, offering guidance for practitioners in probabilistic conformal prediction.

Full Conformal Prediction under Covariate Shift

Jun 2025 – Sep 2025

Independent Research Project, advised by Dr. Michele Caprio

- Proposed a full conformal prediction algorithm based on importance weighted kernel density estimation which provides robustness to specified classes of covariate shift.
- Investigated the use of intermediate representation spaces to render the algorithm computationally tractable in high dimensions while maintaining informative prediction sets.
- Analysed how the choice of intermediate mapping affects set-size efficiency, identifying conditions under which optimality results transfer to the intermediate representation space.
- Developing the resulting manuscript into a preprint, with an early draft available upon request.

Solving the Rubik's Cube with Deep Reinforcement Learning

Sep 2021 – Jan 2022

Early Research Project

- Recreated the Rubik's Cube solving algorithm from Agostinelli et al. (2019), implementing deep reinforcement learning and search methods to achieve comparable performance on low scramble numbers.
- Trained and optimised the neural network policy without the use of machine learning libraries, developing custom backpropagation and search routines to strengthen understanding of underlying principles.
- Developed a C# application visualising the solving process through an interactive 3D animation.

Academic Awards & Scholarships

- Manchester Science and Engineering Excellence Scholarship (£5,000, 2022–2025), awarded in recognition of exceptional academic performance.

Teaching Experience

Oxford Quantum Information Society Seminar

Oct 2025

- Invited to deliver a technical seminar on the interplay between quantum machine learning, distribution drift, and robust uncertainty quantification.
- Presented to an interdisciplinary audience of physicists and mathematicians, communicating advanced concepts with mathematically rigorous examples and clear motivation.

Rock Climbing Coach

Sep 2020 – Sep 2022

- Coached academy athletes in pursuit of national-level competition goals, providing structured training plans and individual performance feedback.
- Led weekly sessions for groups of over 20 climbers, adapting instruction to widely varying experience and ability levels.

Professional Experience

Thule Media

Jun 2024 – Sep 2024

Junior Python Programmer

Chorley, UK

- Automated the analysis of historical financial statements, identifying key transactions and saving approximately £10,000.
- Applied Latent Dirichlet Allocation (LDA) to categorise search queries and identify those relevant to client products.
- Developed software to identify underperforming product categories and refine client keyword targeting from search data.

Turner & Townsend

Jul 2023 – Apr 2024

Summer Intern | Junior Consultant (Part-time, Remote)

Dublin, Ireland

- Joined the Dublin Data Team as a summer intern and was retained part-time during university term-time following strong performance.
- Analysed schedule data and maintained data quality for a major national clean energy infrastructure programme.
- Led discussions with key stakeholders to develop reporting tools improving programme spending visibility.

Technical Skills

- Programming:** Python, C++, SQL, Java, Haskell.
- Machine Learning:** PyTorch, NumPy, SciPy, Matplotlib; experience implementing and evaluating ML models, including uncertainty-aware methods.
- Research Tools:** Git, Linux, LaTeX, VS Code.

Extracurricular

- GB Climbing:** Selected for the Great Britain National Development Squad (2021) for competition climbing.
- Sports Scholarship:** Awarded The University of Manchester Sports Scholarship (£6,000, 2022–2025).